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autoMITRE: AI-Powered Threat Intelligence and Mitigation System

Graduation Project

A Thesis submitted in partial fulfillment of the requirements of
B.Sc. in "CyberSecurity."

by

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Table of Contents

ABSTRACT **3**

ACKNOWLEDGMENT **4**

LIST OF FIGURES **6**

LIST OF TABLES **7**

ABBREVIATIONS ..... ERROR! BOOKMARK NOT DEFINED.

SYMBOLS ..... ERROR! BOOKMARK NOT DEFINED.

CHAPTER 1 - INTRODUCTION **1**

1.1 Motivation *1*
1.2 Problem Statement *2*
1.3 Objectives *4*
1.4 Problem Complexity *2*
1.5 Constraints *6*
1.6 Standards *6*
1.7 Feasibility Study and Business Canvas *7*
1.8 Thesis Organization *8*

CHAPTER 2 - BACKGROUND **3**

2.1 Add sections as required. ..... ***Error! Bookmark not defined.***

CHAPTER 3 - RELATED WORK AND SIMILAR SYSTEMS .. ERROR! BOOKMARK NOT DEFINED.

3.1 Related Work *7*
3.2 Similar Systems *8*

CHAPTER 4 - ANALYSIS **11**

SOFTWARE REQUIREMENTS SPECIFICATIONS IEEE 830 ..... **11**

CHAPTER 5 - DESIGN **20**

5.1 Design Alternatives *20*
5.2 UML Diagrams *24*
5.3 Technologies and Tools Used *32*
5.4 Prototype *35*

CHAPTER 6 - IMPLEMENTATION **40**

6.1 Implementation Environment *40*
6.2 Add sections as required. *40*
6.3 Results and Discussion *40*

CHAPTER 7 - TESTING **41**

SOFTWARE TEST PLAN IEEE 829 ..... **41**

CHAPTER 8 - CONCLUSION AND FUTURE WORK **42**

REFERENCES **42**

APPENDIX A - SENIOR PROJECT SUMMARY REPORT **43**

Abstract

The rapid evolution of cyber threats has created an urgent demand for intelligent, automated, and unified cybersecurity solutions. Traditional defensive mechanisms, although effective within their scope, often operate in isolation and rely heavily on manual interpretation. Cybersecurity analysts depend on multiple specialized tools such as Microsoft Threat Modeling Tool (TMT), OWASP Threat Dragon, IriusRisk, VirusTotal, and Security Information and Event Management (SIEM) systems to identify and mitigate vulnerabilities. However, these tools lack integration and contextual awareness, leading to fragmented analyses, duplicated efforts, and slower incident response.

This project introduces **AutoMITRE — an AI-Powered Threat Intelligence and Mitigation System**, designed to unify, automate, and enhance the threat analysis process across multiple data sources and frameworks. AutoMITRE accepts diverse input formats, including textual threat descriptions, JSON exports from threat modeling tools, PCAP network captures, malware hashes, and system or application logs. Using Artificial Intelligence (AI) and Natural Language Processing (NLP), the system semantically analyzes these inputs to detect threats, categorize attack behaviors, and correlate them with industry-standard frameworks such as **MITRE ATT&CK**, **MITRE D3FEND**, **NIST 800-53**, and **OWASP ASVS**.

Unlike conventional tools, AutoMITRE goes beyond static analysis. It employs an **adaptive learning and predictive intelligence model** that studies historical patterns to forecast future threat trends and potential attack vectors. The system then generates **explainable, step-by-step mitigation strategies**, providing cybersecurity professionals with actionable recommendations that align with standardized defense frameworks.

AutoMITRE further integrates seamlessly with existing security ecosystems by exporting results in **SIEM-compatible formats** (JSON, CSV, and STIX 2.1). This allows the intelligence it produces to be directly ingested by platforms such as **Splunk, QRadar, Elastic Security, and Wazuh**, enabling proactive defense, automated correlation, and improved situational awareness.

By merging threat modeling, AI-driven analytics, and multi-framework mapping within a single platform, AutoMITRE reduces manual workload, enhances accuracy, and transforms fragmented data into actionable, predictive intelligence. This research aims to bridge the existing gap between static threat analysis and dynamic, AI-enhanced mitigation — paving the way for future intelligent cybersecurity ecosystems capable of **anticipating threats before they occur**.

Acknowledgment

The successful completion of this project would not have been possible without the invaluable support, guidance, and encouragement of several individuals and institutions to whom we owe our deepest gratitude.

First and foremost, we would like to express our profound appreciation to our supervisor, **Dr. Ahmed Maher Moustafa**, for his exceptional guidance, insightful advice, and unwavering support throughout every stage of the project. His expertise in cybersecurity and research direction played a vital role in shaping the vision and execution of this work.

We extend our sincere thanks to the **College of Computing and Information Technology (CCIT)** at the **Arab Academy for Science, Technology and Maritime Transport (AASTMT)** for providing a strong academic foundation, technical resources, and a collaborative research environment that made this project possible.

Special thanks are due to our professors, teaching assistants, and peers who contributed valuable feedback, shared knowledge, and helped refine the technical and analytical aspects of this research. Their input significantly enriched the quality of the project and enhanced our understanding of advanced cybersecurity principles.

We would also like to acknowledge the open-source and cybersecurity research communities whose tools, datasets, and frameworks — including MITRE, OWASP, and NIST — formed the backbone of this study. Their commitment to global collaboration and transparency in cybersecurity inspired the development of this project.

Finally, we express our heartfelt gratitude to our families and friends for their constant motivation, patience, and encouragement throughout our academic journey. Their belief in us has been a continuous source of strength and determination.

To everyone who contributed, directly or indirectly, to the success of **AutoMITRE**, we extend our deepest thanks and appreciation.

List of Figures

Figure 1: Enhanced Entity Relationship Diagram (EERD)... **Error! Bookmark not defined.**
Figure 2: Use Case Diagram..... **Error! Bookmark not defined.**
Figure 3: Activity Diagram..... 26
Figure 4: Class diagram..... 27
Figure 5: Data Flow Diagram Level 0..... 28
Figure 6: Data Flow Diagram Level 1..... 29
Figure 7: Data Flow Diagram Level 2..... 29
Figure 8: Sequence Diagram..... 30
Figure 9: State Machine Diagram 31
Figure 10: Flowchart Diagram..... 31

List of Tables

Table 1: Standards table 6
Table 2: comparison of similar systems 10

Chapter 1 - Introduction

In recent years, the cybersecurity landscape has undergone a dramatic transformation, characterized by the rapid emergence of advanced and persistent cyber threats targeting organizations of all sizes. As the attack surface expands through cloud computing, IoT, and interconnected digital systems, traditional manual approaches to threat analysis and mitigation have become increasingly inadequate. Security analysts now face an overwhelming amount of data generated by various tools and platforms, often leading to delayed responses and overlooked vulnerabilities. The growing complexity of these threats highlights the need for intelligent, automated systems capable of not only identifying and analyzing risks from multiple sources but also providing actionable, framework-aligned mitigation strategies. This pressing need for unified, AI-driven cybersecurity analysis forms the foundation and motivation for developing **AutoMITRE**, an advanced platform designed to bridge the gap between threat modeling, intelligence correlation, and proactive defense.

1.1 Motivation

In today's digital age, cyber threats are evolving faster than the tools designed to defend against them. Organizations face a continuous surge of sophisticated attacks ranging from ransomware and phishing to advanced persistent threats (APTs) and insider exploits. The complexity of modern cyberattacks requires multi-layered defense mechanisms supported by continuous threat analysis, intelligence correlation, and proactive mitigation strategies.

To manage this, cybersecurity teams rely on a diverse ecosystem of tools, such as **Microsoft Threat Modeling Tool (TMT)** for system-level threat identification, **OWASP Threat Dragon** and **IriusRisk** for risk modeling, **VirusTotal** for malware intelligence, and **SIEM (Security Information and Event Management)** systems like Splunk, Wazuh, and QRadar for event correlation and alerting.

However, these tools work in **isolation**, each focusing on a single domain. For example:

- **Threat modeling tools** produce design-time reports but lack dynamic analysis.
- **SIEM systems** collect runtime logs but lack predictive and contextual mapping.
- **Malware intelligence platforms** identify known signatures but cannot predict or explain attack trends.

This fragmentation creates a gap in **situational awareness**. Security analysts must manually correlate information between tools, cross-reference multiple frameworks like **MITRE ATT&CK**, **MITRE D3FEND**, and **NIST 800-53**, and derive mitigations manually — a process that is **slow, error-prone, and reactive** rather than proactive.

This challenge motivates the creation of **AutoMITRE**, an AI-powered system that can bridge these gaps by providing **a unified, automated, and intelligent approach** to threat analysis, framework mapping, and mitigation recommendation. By integrating artificial

intelligence and threat modeling principles, AutoMITRE aims to empower cybersecurity teams with **contextual, predictive, and actionable insights** — transforming raw data into meaningful, defense-oriented intelligence.

1.2 Problem Statement

Despite the wide availability of cybersecurity tools, organizations continue to suffer from delayed incident response and poor threat correlation due to **tool fragmentation and lack of automation**. Current workflows depend heavily on human analysts to:

- Collect and interpret threat modeling reports.
- Analyze network traffic or PCAP files.
- Manually map findings to frameworks like MITRE ATT&CK or OWASP ASVS.
- Search for relevant mitigations or defensive strategies.

These manual efforts are not only time-consuming but also fail to scale with the volume and speed of modern threats.

Moreover, existing tools are **reactive** — they only describe what has already occurred. Few systems are capable of **predicting future threats or attack paths** based on patterns found in existing data.

Hence, the central research and engineering problem can be stated as follows:

There is currently no unified, AI-based platform that can automatically analyze diverse cybersecurity inputs, map them to multiple frameworks, generate explainable mitigation strategies, and predict future threats — all while producing outputs compatible with SIEM tools for automated defense workflows.

This problem represents both a **technical challenge** (in data processing, NLP, and AI correlation) and a **practical gap** in real-world cybersecurity operations.

1.3 Problem Complexity

The complexity of the **AutoMITRE** project lies in its multidisciplinary nature, integrating artificial intelligence, natural language processing, cybersecurity frameworks, and SIEM (Security Information and Event Management) interoperability into a single unified system. The platform must handle highly diverse input data, perform intelligent interpretation, generate contextual mappings across multiple standards, and produce outputs that can seamlessly integrate into existing enterprise security environments.

Key challenges and complexities include:

1.3.1 Multi-Format Data Ingestion

AutoMITRE must efficiently process and normalize multiple types of data inputs — including structured reports (JSON, XML), semi-structured logs, unstructured textual threat descriptions, PCAP network captures, and malware hashes.
Each format presents unique parsing and normalization challenges. For instance, JSON threat models from IriusRisk contain structured entities like components and attack surfaces, while raw PCAP files require packet analysis and extraction of behavioral indicators. Designing a unified parser and normalizer capable of handling these heterogeneous formats is a core technical challenge.

1.3.2 Semantic Understanding and Threat Interpretation

The NLP component must accurately comprehend cybersecurity language, terminologies, and contextual relationships. Terms such as “cross-site scripting,” “SQL injection,” or “command and control beaconing” carry domain-specific semantics that cannot be understood through traditional language models.
AutoMITRE must use domain-tuned NLP models capable of extracting attack intent, affected assets, and potential impact from unstructured or partially structured inputs. This requires extensive pre-processing, tokenization, and fine-tuning of AI models on cybersecurity corpora.

1.3.3 Framework Interoperability and Multi-Standard Mapping

One of AutoMITRE’s most significant technical challenges is ensuring interoperability between different cybersecurity frameworks. A single identified threat may correspond to multiple entities across different standards — for example:

- A detected “Privilege Escalation” event aligns with **MITRE ATT&CK (T1068)**, **NIST 800-53 (AC-6)**, and **OWASP ASVS (V4.0.1)**.
  The mapping engine must therefore maintain a dynamic knowledge base that can automatically correlate AI-detected threat categories with relevant controls, mitigations, and references across all frameworks simultaneously. Ensuring consistency and accuracy across these mappings is non-trivial and requires continuous validation against evolving framework versions.

1.3.4 Predictive Analysis and Threat Forecasting

AutoMITRE aims to incorporate an adaptive learning engine capable of identifying **patterns in recurring threats** and predicting **future attack vectors or high-risk scenarios**. Achieving accurate prediction requires continuous model retraining on validated feedback data, ensuring that the system learns from historical incidents and emerging threat

intelligence feeds. This introduces complexity in model design, evaluation metrics, and data labeling — all while maintaining security and privacy of sensitive threat data.

1.3.5 SIEM Correlation and Integration

A major innovation — and complexity — of AutoMITRE lies in its **seamless integration with SIEM platforms**.
The system must produce standardized and enriched threat intelligence outputs that can be directly consumed by SIEM tools such as Splunk, QRadar, or Wazuh.
This requires:

- **Structured export formats** such as STIX 2.1 and JSON for ingestion compatibility.
- **Field normalization** to align with SIEM event schemas (timestamp, source, severity, confidence, etc.).
- **Threat correlation logic**, allowing SIEMs to match AutoMITRE's AI-derived intelligence with live network or endpoint events.
- **Automation triggers**, enabling proactive defense actions (e.g., alert prioritization, rule generation, or playbook execution).

1.3.6 Summary

In summary, AutoMITRE's complexity emerges from its need to act as an **intelligent bridge** between fragmented cybersecurity tools and enterprise security ecosystems. It combines AI-driven semantic reasoning, framework mapping, and SIEM correlation into one cohesive pipeline.
This convergence of artificial intelligence, automation, and cybersecurity standards makes AutoMITRE not only a challenging academic undertaking but also a **strategically valuable contribution** toward the future of integrated and predictive cyber defense.

1.4 Objectives

The overarching goal of the **AutoMITRE project** is to design and develop a comprehensive, AI-powered threat intelligence system that can unify and automate the process of threat detection, framework mapping, and mitigation recommendation.

Specifically, the project aims to:

1. **Accept multi-source input data**, including:
    - Text-based threat descriptions or reports.
    - JSON reports exported from tools such as TMT, IriusRisk, and OWASP Threat Dragon.
    - Network packet captures (PCAP files) for traffic-based threat analysis.
    - System logs or malware hashes for verification through public APIs such as VirusTotal.

2. **Process and interpret data** using Natural Language Processing (NLP) and AI classification models to identify key threat types, indicators, and attack patterns.
3. **Map identified threats** to global cybersecurity frameworks, including **MITRE ATT&CK**, **MITRE D3FEND**, **NIST 800-53**, and **OWASP ASVS**, providing complete context for each finding.
4. **Generate actionable mitigations** aligned with best practices and framework defenses, presented in clear, human-readable steps.
5. **Predict potential future threats or next-stage attack behaviors** based on detected patterns, leveraging AI trend analysis and frequency correlation.
6. **Visualize findings** through an interactive dashboard showing detected threats, framework coverage, and predicted trends.
7. **Export results** in standardized formats such as JSON, CSV, and **STIX 2.1**, enabling integration with **SIEM systems** (Splunk, QRadar, Elastic Security, Wazuh) for further correlation and automated response.

1.5 Constraints

AutoMITRE's design and implementation are subject to the following constraints:

Software Constraints:
AutoMITRE is constrained by the availability, stability, and usage limits of external software components and APIs. Integration with third-party services such as VirusTotal, threat modeling tools, and SIEM platforms depends on API accessibility, rate limits, and licensing restrictions. The system must rely on open-source or academic-licensed AI/ML frameworks, which may limit model complexity and real-time performance. Compatibility with standardized formats (JSON, STIX 2.1, CSV) is mandatory to ensure interoperability with SIEM solutions.

Hardware Constraints:
The project is developed under limited computational resources, restricting large-scale model training and real-time deep packet inspection for high-volume PCAP files. GPU availability is constrained, which affects advanced deep learning experimentation. Storage capacity also limits the retention of large datasets such as PCAP files and historical threat logs.

Time Constraints:
The development timeline is bounded by the academic semester, requiring the project to focus on a functional prototype rather than a fully production-ready system. As a result, only core features such as threat analysis, framework mapping, and visualization are prioritized, while extensive optimization and large-scale validation are deferred.

Cost Constraints:
The project operates under a minimal budget with no allowance for premium APIs, enterprise SIEM licenses, or large cloud infrastructure. All tools, datasets, and services must be free, open-source, or available through educational access, limiting scalability and commercial deployment readiness.

1.6 Standards

To ensure the system's design, documentation, and evaluation adhere to recognized best practices, AutoMITRE follows several **international standards** and frameworks:

Table 1: Standards table

Category	Standard / Framework	Purpose
Software Engineering	**IEEE 830**	Software Requirements Specification (SRS)
	IEEE 829	Software Testing Documentation
	IEEE 12207	Software Development Lifecycle (SDLC)
Cybersecurity Frameworks	**MITRE ATT&CK & D3FEND**	Attack and defense mapping reference models
	NIST 800-53	Security and privacy control catalog
	OWASP ASVS	Web application vulnerability assessment
Threat Intelligence Standards	**STIX / TAXII**	Structured threat intelligence exchange formats

Adhering to these standards ensures consistency, interoperability, and professional quality in

both documentation and design.

1.7 Feasibility Study and Business Canvas

1.7.1 Technical Feasibility

AutoMITRE leverages widely used open-source technologies that are mature and reliable:

- **Python 3.x** as the primary language for AI, data parsing, and integration.
- **spaCy**, **Hugging Face Transformers**, and **scikit-learn** for NLP and classification.
- **Scapy** for PCAP analysis and network feature extraction.
- **Streamlit** for dashboard visualization.
- **SQLite/PostgreSQL** for data management.
- **stix2** library for exporting STIX 2.1 compatible files.
  All of these are freely available and well-documented, making the system technically achievable by a student team.

1.7.2 Operational Feasibility

The system is designed with modularity in mind — each of the six team members can focus on a separate module:
Input parsing, AI/NLP engine, framework mapping, predictive analysis, visualization, and export integration.
This division ensures balanced workload and parallel development.

1.7.3 Economic Feasibility

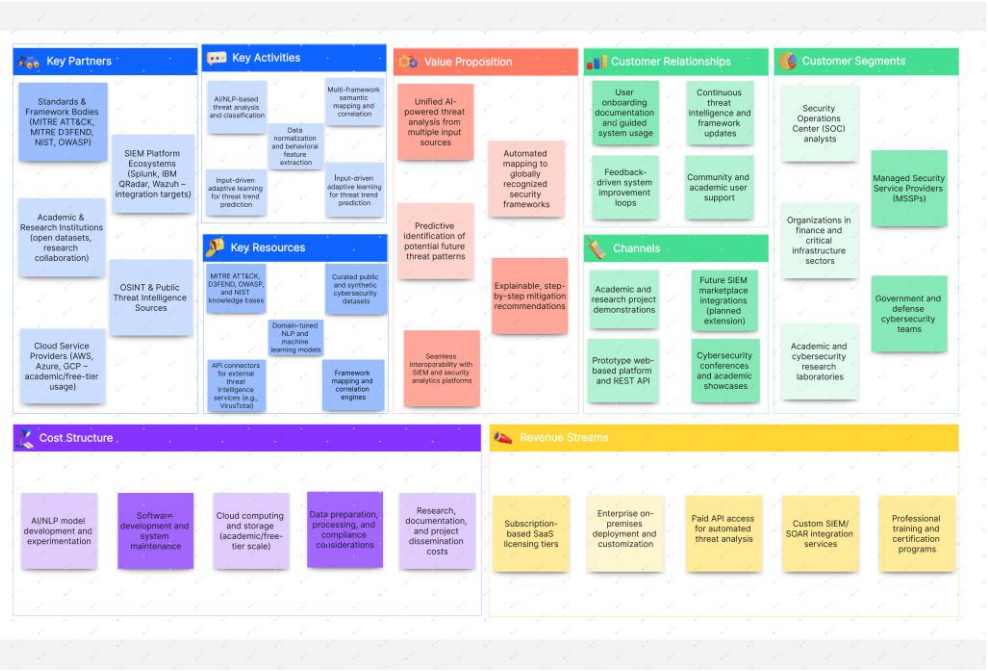
The system uses only open-source components and publicly accessible datasets, eliminating licensing costs. The main resource investment is time and effort.

1.7.4 Schedule Feasibility

The project is separated into two parts, project 1 to be submitted in the first semester and project 2 to be submitted in the second semester.

1.7.5 Academic Feasibility

The project integrates key learning outcomes from cybersecurity, artificial intelligence, and software engineering courses. It demonstrates both theoretical understanding and practical application, making it suitable as a senior graduation project.



1.8 Thesis Organization

The remainder of this thesis is organized as follows:

- **Chapter One** introduces the project motivation, problem statement, objectives, constraints, standards, and feasibility.
- **Chapter Two** provides a detailed background on cybersecurity concepts, frameworks, AI/NLP technologies, and SIEM systems.
- **Chapter Three** presents a comparative analysis of existing tools and discusses related work in threat modeling and automation.
- **Chapter Four** defines the system's requirements and use cases according to IEEE 830 standards, including data flow and user interaction diagrams.
- **Chapter Five** outlines the system's architectural design, major modules, technologies, and high-level data processing flow.
- **Chapter Six and beyond** will discuss implementation, testing, results, and conclusions.

Chapter 2 – Background and Similar Systems

2.1 Background

The cybersecurity domain has evolved significantly in recent years due to the continuous rise of complex, multi-stage, and automated attacks targeting both individuals and organizations. With increasing digitalization, the attack surface has expanded, encompassing networks, cloud infrastructures, IoT devices, and industrial control systems.

Traditional defense mechanisms — such as signature-based antivirus software and rule-based intrusion detection systems — are no longer sufficient to detect and prevent sophisticated cyber threats. These conventional tools often fail to recognize new or modified attack techniques, which require contextual analysis and correlation across multiple data sources.

The introduction of **Artificial Intelligence (AI)** and **Machine Learning (ML)** into cybersecurity has brought a paradigm shift, enabling systems to automatically learn from data, identify anomalies, and adapt to evolving threats. However, most AI-based tools are developed for specific domains (e.g., network intrusion detection, malware classification, or behavioral analysis) and lack interoperability and unified threat context.

The **AutoMITRE** project addresses this limitation by designing a unified, AI-driven system that integrates multi-source threat intelligence, framework mapping, and mitigation generation within a single automated platform. This chapter provides the background knowledge necessary to understand the technologies, frameworks, and concepts that underpin the AutoMITRE system.

2.1.1 Cyber Threat Modeling

Threat modeling is the process of identifying potential threats, vulnerabilities, and attack paths that could compromise a system. It enables security teams to proactively analyze how an adversary might exploit weaknesses before an incident occurs.

Common methodologies for threat modeling include:

- **STRIDE (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege):** Used by Microsoft's Threat Modeling Tool (TMT) to categorize threats based on attacker goals.
- **DREAD:** Quantifies threat risk levels using Damage, Reproducibility, Exploitability, Affected Users, and Discoverability.
- **PASTA (Process for Attack Simulation and Threat Analysis):** A risk-centric approach combining business impact with technical vulnerabilities.

- **OCTAVE (Operationally Critical Threat, Asset, and Vulnerability Evaluation):** Focused on organizational-level risk assessment.

While these models are effective individually, their outputs are tool-specific and require manual interpretation. The lack of integration between these tools and modern AI-driven analysis systems often results in disjointed workflows. AutoMITRE aims to automate this process by taking threat model outputs and converting them into machine-understandable data for AI-based analysis and cross-framework mapping.

2.1.2 Cybersecurity Frameworks and Standards

The security community relies on a variety of global frameworks to standardize threat classification and mitigation. AutoMITRE incorporates several of these frameworks to provide context-aware intelligence.

1- MITRE ATT&CK Framework

The **MITRE ATT&CK (Adversarial Tactics, Techniques, and Common Knowledge)** framework is a globally recognized knowledge base of adversary tactics and techniques. It documents how attackers operate in real-world scenarios — including persistence, privilege escalation, and data exfiltration.
AutoMITRE uses ATT&CK as a reference model for mapping identified threat behaviors to specific techniques and adversarial patterns, providing contextual insight into each detected event.

2- MITRE D3FEND Framework

MITRE D3FEND is a complementary framework to ATT&CK, focusing on defensive techniques and mitigations. It links each ATT&CK technique with a set of defensive countermeasures.
In AutoMITRE, D3FEND enables the system to automatically recommend relevant mitigations once a threat is identified.

3- NIST SP 800-53

The **National Institute of Standards and Technology (NIST)** provides guidelines for security and privacy controls within information systems. NIST SP 800-53 ensures compliance and regulatory alignment in organizational cybersecurity operations.
AutoMITRE references these controls to validate its recommended mitigations and ensure they adhere to global best practices.

4- OWASP Application Security Verification Standard (ASVS)

The **OWASP ASVS** is an application-level framework that defines a standardized basis for testing web application security. It provides developers and security teams with a clear structure for identifying vulnerabilities.
AutoMITRE uses OWASP ASVS for aligning software-level threats, such as input validation flaws, session management issues, and injection vulnerabilities.

2.1.3 Artificial Intelligence and NLP in Cybersecurity

AI and **Natural Language Processing (NLP)** play a crucial role in automating and enhancing cybersecurity analysis. By enabling systems to process unstructured data — such as reports, logs, or textual descriptions — AI models can uncover hidden relationships and detect anomalies that rule-based systems might miss.

1- Machine Learning for Threat Detection

Machine learning algorithms, such as decision trees, random forests, and neural networks, have been successfully applied in identifying patterns of malicious activity. They analyze large datasets to distinguish normal from abnormal behavior.
In AutoMITRE, ML models are responsible for classifying detected threats and predicting likely future attack trends.

2- Natural Language Processing for Threat Analysis

NLP enables machines to understand cybersecurity terminology and extract key information from textual inputs such as incident reports or alerts.
AutoMITRE uses NLP to:

- Identify attack types, affected components, and severity levels from text-based inputs.
- Match extracted information against framework taxonomies (e.g., ATT&CK and D3FEND).
- Generate summaries and structured representations of unstructured threat data.

2.1.4 Security Information and Event Management (SIEM) Systems

SIEM systems aggregate and analyze log and event data from across an organization's IT environment. They provide real-time monitoring, correlation, and alerting capabilities. Examples include **Splunk Enterprise Security**, **IBM QRadar**, and **Wazuh**.

While SIEM tools are essential for operational monitoring, they rely on structured, well-defined event feeds. Analysts must often manually interpret data and correlate it with threat intelligence feeds or framework references.

AutoMITRE enhances SIEM capabilities by:

- Exporting **STIX 2.1** or **JSON-formatted** intelligence for seamless SIEM ingestion.
- Providing pre-correlated insights that help SIEM systems detect patterns earlier.
- Reducing alert fatigue through AI-based threat prioritization.

This integration allows AutoMITRE to function as both a **threat interpretation engine** and a **SIEM intelligence feeder**, bridging the gap between strategic analysis and operational response.

2.1.5 Adaptive Learning and Threat Prediction

AutoMITRE's **Adaptive Learning Module** uses feedback-driven AI to improve accuracy and predictive capability over time. By analyzing trends in previously detected threats, it can forecast emerging attack techniques and highlight areas of potential risk before exploitation occurs.
This predictive capability is essential for transforming traditional reactive defenses into **proactive cybersecurity ecosystems** capable of evolving with the threat landscape.

2.1.6 Summary Of Background

This chapter has discussed the theoretical and technological background behind the AutoMITRE project, including the foundations of threat modeling, the roles of global security frameworks, and the integration of AI and NLP in modern cybersecurity systems. AutoMITRE builds upon these principles to create a unified, intelligent, and extensible system capable of understanding, mapping, and mitigating cybersecurity threats — all while integrating seamlessly with existing security infrastructure such as SIEM platforms.

2.2 Related Work and Similar Systems

Cybersecurity threat modeling and intelligence analysis have evolved significantly with the advancement of artificial intelligence and data analytics. Many tools and research works aim to help organizations identify, assess, and mitigate threats, yet most of them operate independently and lack unified integration.

This chapter reviews the **existing related work** and **similar systems** in the field of threat modeling and cyber defense automation. It also highlights the **research and functional gap** that AutoMITRE aims to address — the absence of an AI-powered platform that can analyze various input sources, automatically map threats to global frameworks such as MITRE ATT&CK and D3FEND, and provide actionable mitigation strategies in a single, user-friendly environment.

2.2.1 Related Work

Recent research in cybersecurity has focused on enhancing automation and intelligence in threat analysis using AI and framework-based mapping. Below is a summary of relevant literature and tools that influence AutoMITRE's design.

1. **AI-Assisted Threat Analysis**
   Several studies have introduced AI and Natural Language Processing (NLP) models to analyze unstructured threat reports. These systems can extract attack indicators or behaviors from text but are often limited to one data type (e.g., threat intelligence feeds). AutoMITRE expands on this concept by accepting multiple input formats (text, JSON, PCAP, and logs) and using NLP for broader detection and framework mapping.
2. **MITRE ATT&CK and D3FEND Framework Integration**
   Research has shown that the MITRE ATT&CK matrix provides a standardized vocabulary for attacker behaviors, while MITRE D3FEND focuses on defensive techniques. However, most tools utilize these frameworks for manual referencing. AutoMITRE introduces automated framework correlation and AI-based threat-to-technique mapping.
3. **Threat Modeling and Risk Assessment Tools**
   Academic and industrial tools such as Microsoft TMT, OWASP Threat Dragon, and IriusRisk support structured threat modeling based on known methodologies (STRIDE, PASTA, etc.). These tools, however, lack adaptive intelligence or cross-framework mapping capabilities. AutoMITRE aims to bridge this gap by combining static model outputs with AI-driven contextual analysis.
4. **Integration with SIEM Systems**
   Prior research on Security Information and Event Management (SIEM) platforms emphasizes data correlation and alert prioritization. Still, SIEMs depend on structured input and often miss semantic relationships between events. AutoMITRE complements these systems by exporting enriched intelligence in standardized formats (JSON/STIX) for direct SIEM ingestion.

Summary:
While existing research focuses on either AI-driven detection or framework-based documentation, **none combine AI, framework mapping, and predictive analysis in a unified educational system**. AutoMITRE fills this research and functionality gap.

2.2.2Similar Systems

This section compares existing tools and systems similar to AutoMITRE, outlining their functions, inputs/outputs, technologies, and limitations.

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1. Microsoft Threat Modeling Tool (TMT)

- **Description:** Developed by Microsoft, TMT allows users to create data flow diagrams (DFDs) and identify threats using the STRIDE model.
- **Main Function:** Identifies potential threats and vulnerabilities in software design.
- **Input / Output:** Data Flow Diagram → Threat report (CSV or text).
- **Technologies Used:** .NET Framework, STRIDE methodology.
- **Results:** Generates categorized lists of potential threats.
- **Advantages:** Easy to use, structured risk assessment.
- **Disadvantages:** No AI automation, no mapping to ATT&CK/D3FEND, and lacks mitigation suggestions.

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2. OWASP Threat Dragon

- **Description:** An open-source threat modeling tool that focuses on web and application security.
- **Main Function:** Provides graphical threat modeling using the STRIDE approach.
- **Input / Output:** Diagram input → Threat analysis report.
- **Technologies Used:** JavaScript, Node.js, Electron.
- **Results:** Visual representation of threats and mitigations.
- **Advantages:** Free, cross-platform, integrates with OWASP standards.
- **Disadvantages:** No AI component, no multi-framework correlation.

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3. IriusRisk

- **Description:** A commercial threat modeling and risk management tool that automates parts of the threat analysis process.
- **Main Function:** Generates threats and countermeasures based on architecture inputs.
- **Input / Output:** System architecture diagrams → Threat and control reports.
- **Technologies Used:** Java, REST API, integrated rule-based engine.
- **Results:** Suggests mitigations based on predefined templates.
- **Advantages:** Automates threat generation and control mapping.

- **Disadvantages:** Closed-source, limited free version, not AI-powered.

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4. VirusTotal

- **Description:** Online platform for scanning files and URLs for malware using multiple antivirus engines.
- **Main Function:** Detects known malicious files or domains.
- **Input / Output:** File/URL → Analysis report (JSON).
- **Technologies Used:** API-based service, antivirus engine aggregation.
- **Results:** Provides detection ratio and related threat metadata.
- **Advantages:** Reliable detection of known threats.
- **Disadvantages:** No behavioral mapping or framework integration.

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5. Splunk Enterprise Security (SIEM)

- **Description:** A commercial SIEM solution for large-scale event collection and threat correlation.
- **Main Function:** Aggregates and analyzes security events from logs and network traffic.
- **Input / Output:** Log files, network data → Alerts and dashboards.
- **Technologies Used:** Machine learning models, data indexing engine.
- **Results:** Detects anomalies and prioritizes security events.
- **Advantages:** Real-time detection, scalability.
- **Disadvantages:** Not designed for threat modeling or AI-driven mitigations.

2.2.3 Comparison Summary

Table 2: comparison of similar systems

System	AI/NLP	Framework Mapping	Mitigation Generation	SIEM Integration	Open Source
Microsoft TMT	✗	✓ STRIDE only	✗	✗	✓
OWASP Threat Dragon	✗	✓ STRIDE only	Limited	✗	✓
IriusRisk	✓ (Rule-based)	✓ STRIDE/NIST	(Template)	(API)	✗
VirusTotal	✓ (Detection only)	✗	✗	✓ (API)	✓
Splunk SIEM	✓ (Behavioral ML)	✗	✗	✓	✗
AutoMITRE (Proposed)	✓ (AI + NLP)	✓ (ATT&CK, D3FEND, OWASP, NIST)	✓ (Dynamic AI-generated)	✓ (STIX/JSON Export)	✓

2.2.4 autoMITRE contribution

AutoMITRE combines and extends the capabilities of the tools above by:

- Integrating multiple frameworks (ATT&CK, D3FEND, OWASP, NIST) into one unified AI-based system.
- Allowing multi-format inputs (text, PCAP, JSON, hashes).
- Automatically generating mitigation strategies and exportable SIEM-ready reports.
- Providing adaptive learning that predicts potential future threats.

In summary, while existing tools specialize in either **threat modeling** or **event monitoring**, AutoMITRE provides a **bridge between the two** — enabling both **academic learning** and **automated framework-based defense recommendations**.

Chapter 4 - Analysis

Software Requirements Specifications IEEE 830

This chapter presents a detailed analysis of the proposed **AutoMITRE** system. It focuses on translating the problem statement and project objectives into clear, structured system requirements and analytical models that define how the system operates and interacts with its users and external platforms.

The analysis begins by identifying the key stakeholders and user roles, followed by a breakdown of the functional requirements that describe the core capabilities of AutoMITRE, such as threat input acquisition, AI-driven threat detection, framework mapping, mitigation generation, prediction, and integration with SIEM systems. Each requirement is prioritized to reflect its importance to the system's core functionality and real-world applicability.
In addition, this chapter outlines the non-functional requirements, including performance, scalability, availability, security, usability, and interoperability, which are essential for evaluating the system's quality and feasibility. Since AutoMITRE has not yet been deployed, appropriate simulation and evaluation tools are identified to estimate and validate these requirements.

Overall, this analysis chapter serves as the foundation for the system design and implementation phases, ensuring that AutoMITRE is technically sound, aligned with cybersecurity best practices, and capable of addressing the challenges faced by modern Security Operations Centers (SOCs).

4.1 Functional Requirements (FRs)

These describe **what the system must do**.

FR1. User Authentication & Access Control

Priority: MEDIUM

FR1.1 The system **should** allow users to create accounts and authenticate securely.
FR1.2 The system **should** support role-based access control (Analyst, Administrator).
FR1.3 The system **should** restrict sensitive operations (e.g., SIEM export, system configuration) to authorized roles only.

Rationale: Required for production readiness, not essential for core analysis logic.

FR2. Input Acquisition and Data Upload

Priority: HIGH

FR2.1 The system **shall** accept multiple input formats, including text, JSON, PCAP files, log files, and file hashes.
FR2.2 The system **shall** provide a web-based interface for uploading input data.
FR2.3 The system **shall** support importing threat modeling reports (e.g., JSON/XML) from tools such as TMT, IriusRisk, and Threat Dragon.
FR2.4 The system **shall** be capable of fetching input data directly from external intelligence APIs (e.g., VirusTotal, TMT API).

Rationale: Without input acquisition, the system cannot operate.

FR3. Input Parsing & Normalization

Priority: HIGH

FR3.1 The system **shall** parse and normalize all supported inputs into a unified internal data schema.
FR3.2 The system **shall** validate incoming data and notify users of malformed or unsupported inputs.

Rationale: All downstream intelligence depends on normalized data.

FR4. Threat Detection Engine

Priority: HIGH

FR4.1 The system **shall** use AI/NLP techniques to extract threats from text-based inputs.
FR4.2 The system **shall** perform traffic and feature extraction from PCAP files and log data.
FR4.3 The system **shall** classify detected threats and assign severity and confidence scores.
FR4.4 The system **shall** retrieve malware reputation data from external intelligence services (e.g., VirusTotal) when file hashes are provided.

Rationale: This represents the core analytical capability of the system.

FR5. Framework Mapping

Priority: HIGH

FR5.1 The system **shall** map detected threats to MITRE ATT&CK techniques.
FR5.2 The system **shall** map threats to MITRE D3FEND defensive techniques.
FR5.3 The system **shall** map applicable threats to OWASP ASVS requirements.
FR5.4 The system **shall** map threats to relevant NIST SP 800-53 security controls.
FR5.5 The system **shall** persist all framework mappings within the Threat Mapping data store.

Rationale: Framework correlation is the primary differentiator of the project.

FR6. Mitigation Plan / Steps Generation

Priority: HIGH

FR6.1 The system **shall** automatically generate mitigation steps based on mapped security frameworks.
FR6.2 The system **shall** prioritize mitigation recommendations according to threat severity and potential impact.
FR6.3 The system **shall** store generated mitigation plans in the Mitigation data repository.

Rationale: Analysts require actionable guidance, not just detection results.

FR7. Threat Prediction Module

Priority: MEDIUM

FR7.1 The system **should** analyze historical input data to identify recurring threat patterns.
FR7.2 The system **should** predict potential future threats that the organization may encounter.
FR7.3 The system **should** store prediction outputs for trend analysis and reporting.

Rationale: Innovative capability that enhances the system but depends on prior modules.

FR8. Output Generation & Visualization

Priority: HIGH

FR8.1 The system **shall** present detected threats, framework mappings, mitigations, and predictions through an interactive dashboard.
FR8.2 The system **shall** generate downloadable reports in formats such as PDF and JSON.
FR8.3 The system **shall** provide visual risk representations, including heatmaps and coverage summaries.

Rationale: Results must be clearly interpretable by analysts.

FR9. SIEM Integration

Priority: MEDIUM

FR9.1 The system **should** support exporting analysis results in SIEM-compatible formats (e.g., JSON, CSV, STIX 2.1).
FR9.2 The system **should** log all export operations for auditing and traceability purposes.

Rationale: Important for enterprise environments but not essential for core functionality.

4.2 Non-Functional Requirements (NFRs)

NFR1 — Performance Requirements

Description:
The system shall analyze uploaded inputs (PCAP files, logs, text, and reports) and produce threat analysis, framework mapping, and mitigation results within an acceptable response time.

Estimated Metrics (Simulated):

- Small input (logs/text): ≤ **5 seconds**
- Medium input (PCAP < 50MB): ≤ **15 seconds**
- Large input (PCAP 50–200MB): ≤ **30 seconds**
- Throughput: **20–50 analysis requests/hour** (prototype)

Simulation & Evaluation Tools:

- Apache JMeter (load & response time simulation)
- Locust (API-level performance testing)
- Python time profiling (`time`, `cProfile`)

NFR2 — Availability

Description:
The system shall be available for use by analysts during normal operational hours with minimal downtime.

Estimated Metrics (Simulated):

- Target availability: **95–98%**
- Planned downtime only (maintenance)
- Single-instance prototype environment

Simulation & Evaluation Tools:

- Manual fault injection (service stop/restart)
- Docker container restart tests

NFR3 — Scalability

Description:
The system shall support increased input volume and concurrent users by scaling horizontally or vertically in a simulated environment.

Estimated Metrics (Simulated):

- Concurrent users: **5 → 20 analysts**
- Input volume increase: **2× baseline**
- Linear response degradation acceptable

Simulation & Evaluation Tools:

- Locust (concurrent user simulation)
- Docker Compose (service scaling)

NFR4 — Reliability & Fault Tolerance

Description:
The system shall continue functioning correctly in case of partial failures such as invalid inputs, API timeouts, or service crashes.

Estimated Metrics (Simulated):

- Successful recovery from failures: **≥ 90%**
- Graceful error handling without data loss
- No system-wide crash on single module failure

Simulation & Evaluation Tools:

- Chaos testing (manual service interruption)
- API failure mocking (Postman / Mock servers)

NFR5 — Security Requirements

Description:
The system shall protect sensitive threat data, uploaded files, and API credentials from unauthorized access.

Estimated Metrics (Simulated):

- Authentication success rate: **100%**
- No unauthorized data access during testing
- Input sanitization coverage: **≥ 95%**

Simulation & Evaluation Tools:

- OWASP ZAP (basic vulnerability scanning)
- Burp Suite Community Edition

NFR6 — Usability

Description:
The system shall provide an intuitive interface allowing SOC analysts to upload data, view results, and export reports with minimal training.

Estimated Metrics (Simulated):

- Task completion rate: **≥ 90%**
- Average learning time: **< 10 minutes**
- User satisfaction (mock survey): **4/5**

Simulation & Evaluation Tools:

- Figma usability testing
- Heuristic evaluation (Nielsen principles)
- User scenario walkthroughs

NFR7 — Maintainability

Description:
The system shall be modular, well-documented, and easy to update or extend with new frameworks or detection models.

Estimated Metrics (Simulated):

- Code modularity: High (micro-service or layered design)
- Average bug fix time: **< 1 day**
- Documentation coverage: **≥ 80%**

Simulation & Evaluation Tools:

- Git version control metrics
- Static code analysis tools

NFR8 — Interoperability

Description:
The system shall integrate with external tools and SIEM platforms using standard formats and APIs.

Estimated Metrics (Simulated):

- Export format success rate (JSON/STIX/CSV): **100%**
- SIEM compatibility (simulated): **Splunk-like ingestion**
- API response success rate: **≥ 95%**

Simulation & Evaluation Tools:

- Postman (API simulation)
- STIX 2.1 validators
- Mock SIEM ingestion scripts

4.3 Primary Stakeholders

4.3.1 End Users (Cybersecurity Analysts / SOC Analysts)

- Use the system to analyze threats.
- Upload input data (PCAP, logs, text, reports).
- Review detected threats, mitigations, and predictions.

4.3.2 System Administrators

- Manage user accounts and access control.
- Configure SIEM integration & external API keys (VirusTotal, TMT API).
- Maintain system availability and performance.

4.4 Secondary Stakeholders

4.4.1 Security Engineers / Threat Hunters

- Use the output for deeper investigations.
- Validate threat mappings (ATT&CK, D3FEND, OWASP, NIST).
- Apply mitigation recommendations.

4.4.2 Incident Response Teams (IR Teams)

- Benefit from predictions of upcoming threats.
- Use SIEM-enriched outputs during incident handling.

4.4.3 Compliance & Governance Teams

- Use framework-based mappings for:
    - Compliance reporting (NIST 800-53)
    - Security audits
    - Risk assessments

4.5 Technical Stakeholders

4.5.1 Developers + AI engineers (separated)

- Design, implement, test, and maintain the system components.
- Responsible for AI engine, mapping module, parsing engine, and EERD/DFDs.

4.5.2 API Providers (External Systems)

These are stakeholders because AutoMITRE depends on their APIs:

- **VirusTotal API**
- **Microsoft Threat Modeling Tool (TMT) API**
- **IriusRisk API**
- **OWASP Threat Dragon API**
- **MITRE ATT&CK/STIX/TAXII sources**

The stability and accessibility of these services affect project functionality.

4.6 External Integration Stakeholders

4.6.1 SIEM Systems

AutoMITRE exports enriched threat data to:

- Splunk
- IBM QRadar
- ArcSight
- Elastic SIEM
- Microsoft Sentinel

These systems depend on correct formatting (STIX, JSON, CSV) for ingestion.

4.7 Indirect Stakeholders

4.7.1 Organizations Using the Tool

- Benefit from improved security posture.
- Reduce time spent on manual threat analysis.
- Improve detection and response workflows.

4.7.2 Future Developers / Researchers

- Use the project as a baseline for improving:
    - AI-driven threat modeling
    - Framework mapping
    - Automated mitigations
    - Threat prediction

Chapter 5 - Design

Engineering design is a process of devising a system, component, or process to meet desired needs and specifications within constraints. It is an iterative, creative, decision-making process in which the basic sciences, mathematics, and engineering sciences are applied to convert resources into solutions. Engineering design involves identifying opportunities, developing requirements, performing analysis and synthesis, generating multiple solutions, evaluating solutions against requirements, considering risks, and making trade-offs, for the purpose of obtaining a high-quality solution under the given circum- stances. For illustrative purposes only, examples of possible constraints include accessibility, aesthetics, codes, constructability, cost, ergonomics, extensibility, functionality, interoperability, legal considerations, maintainability, manufacturability, marketability, policy, regulations, schedule, standards, sustainability, or usability.

5.1 Design Alternatives

During the design phase of the AutoMITRE system, multiple architectural and technical alternatives were evaluated to determine the most suitable approach that balances accuracy, scalability, implementation complexity, and academic feasibility. This section discusses the main design alternatives considered and justifies the final design decisions.

5.1.1 Input Handling Design Alternatives

Alternative 1: Single Input Type (Threat Modeling Reports Only)
This alternative restricts the system input to threat modeling tool outputs (e.g., Microsoft TMT reports).

Advantages:
- Simplified parsing and processing
- Lower implementation complexity

Disadvantages:
- Limited applicability
- Does not support real-world SOC workflows
- Reduces system flexibility

Alternative 2: Multi-Source Input Handling (Selected Design)

The system accepts multiple input types such as PCAP files, logs, text-based threats, threat modeling reports, and malware hashes.

Advantages:

- Supports diverse cybersecurity scenarios
- Aligns with real SOC analyst workflows
- Enhances system usability and extensibility

Disadvantages:

- Higher parsing and processing complexity
- Selected Approach: Multi-source input handling to maximize system applicability and intelligence coverage.

5.1.2 Threat Analysis Approach

Alternative 1: Rule-Based Analysis Only
Threat detection relies solely on predefined rules and signatures.

Advantages:

- High explainability
- Easy validation

Disadvantages:

- Limited adaptability
- Poor detection of unknown or evolving threats

Alternative 2: Machine Learning–Based Analysis Only
Threats are detected using machine learning models trained on datasets.

Advantages:

- Capable of detecting unknown patterns
- Adaptive to new threats

Disadvantages:

- Lower explainability
- Requires large datasets

Alternative 3: Hybrid Rule-Based + ML Approach (Selected Design)
Combines rule-based logic with machine learning and NLP techniques.

Advantages:

- Balanced accuracy and explainability
- Better detection coverage
- Suitable for academic validation

Disadvantages:

- Slightly higher system complexity

- **Selected Approach:** Hybrid analysis model for improved accuracy and interpretability.

5.1.3 Framework Mapping Design

Alternative 1: Single Framework Mapping
Mapping threats to only one framework (e.g., MITRE ATT&CK).

Advantages:
- Simple implementation

Disadvantages:
- Limited defensive and compliance coverage

Alternative 2: Multi-Framework Mapping (Selected Design)
Mapping threats to multiple frameworks including MITRE ATT&CK, MITRE D3FEND, OWASP, and NIST.

Advantages:
- Comprehensive threat and defense visibility
- Supports both offensive and defensive perspectives
- Improves decision-making

Disadvantages:
- Requires correlation logic between frameworks
- Selected Approach: Multi-framework mapping to provide holistic threat intelligence.

5.1.4 External Integration Design

Alternative 1: Manual Upload of External Reports
Users manually upload reports generated by external tools.

Advantages:
- No dependency on APIs
-

Disadvantages:
- Manual effort
- Fragmented workflow

Alternative 2: Direct API Integration (Selected Design)
The system integrates directly with external tools such as VirusTotal and threat modeling tools via APIs.

Advantages:
- Unified workflow
- Reduced manual intervention
- Improved automation

Disadvantages:

- Dependency on API availability and limits
- Selected Approach: API-based integration to unify the threat analysis process.

5.1.5 Data Output & SIEM Integration Design

Alternative 1: Human-Readable Reports Only
Outputs are limited to dashboards and PDF-style reports.

Advantages:

- Easy interpretation

Disadvantages:

- Cannot be reused by other security systems

Alternative 2: Machine-Readable Outputs (Selected Design)
Outputs are generated in SIEM-compatible formats such as JSON, STIX, and CSV.

Advantages:

- Enables automated ingestion by SIEM platforms
- Supports operational security workflows

Disadvantages:

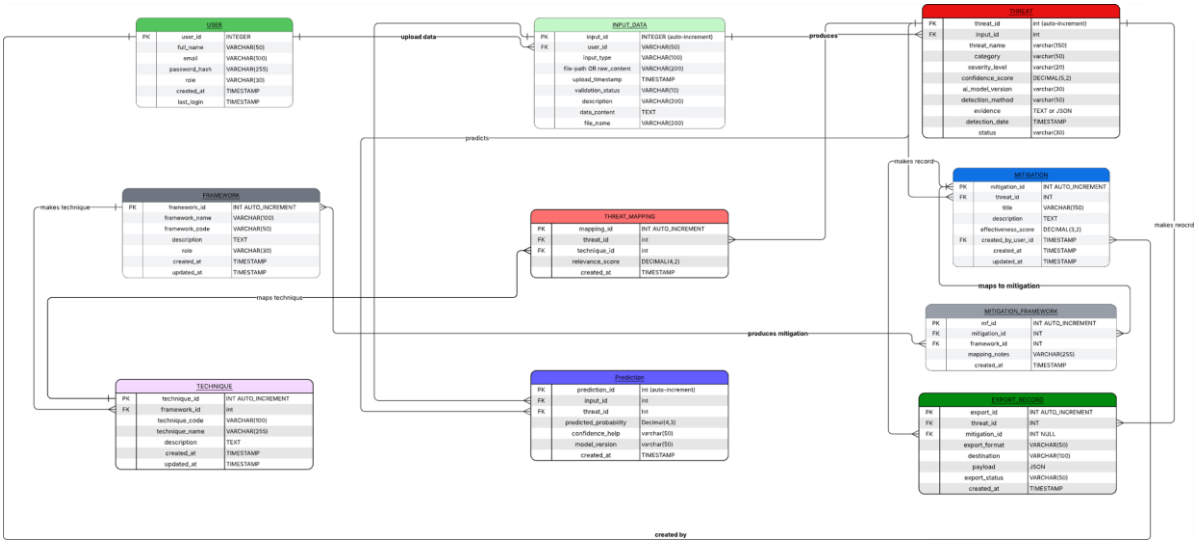
- Requires strict data normalization

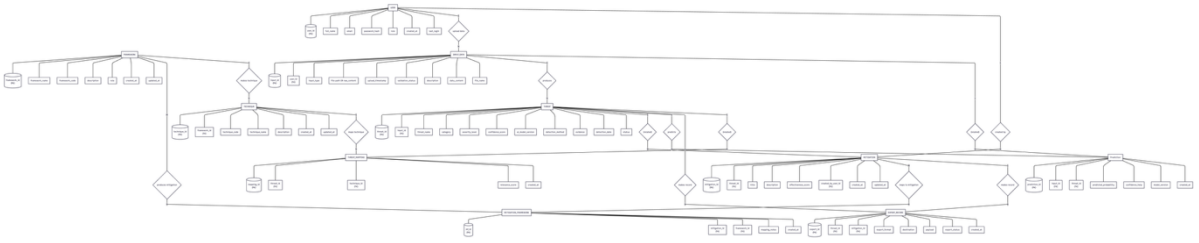
Selected Approach: Machine-readable outputs to maximize interoperability.

5.2 UML Diagrams

5.2.1 Enhanced Entity Relationship Diagram (EERD):

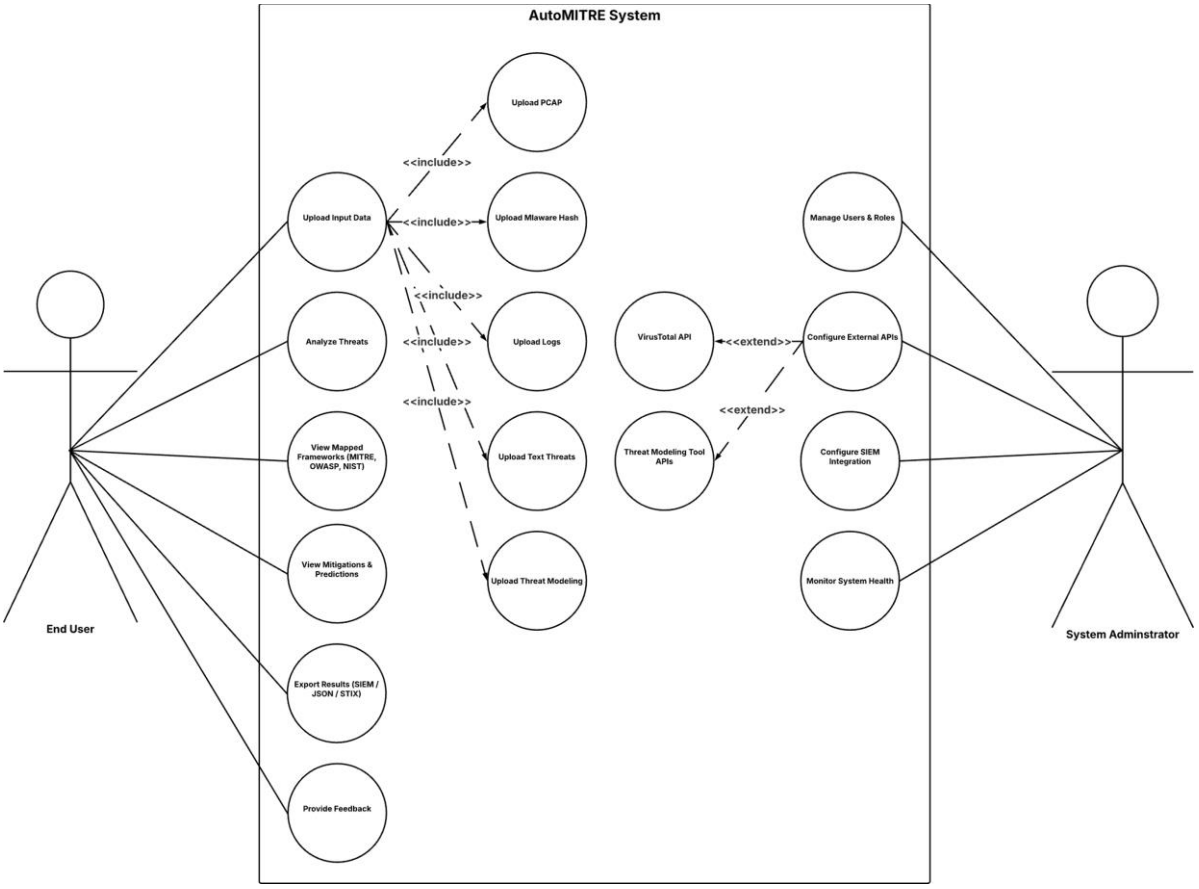
Figure 1: Enhanced Entity Relationship Diagram (EERD)





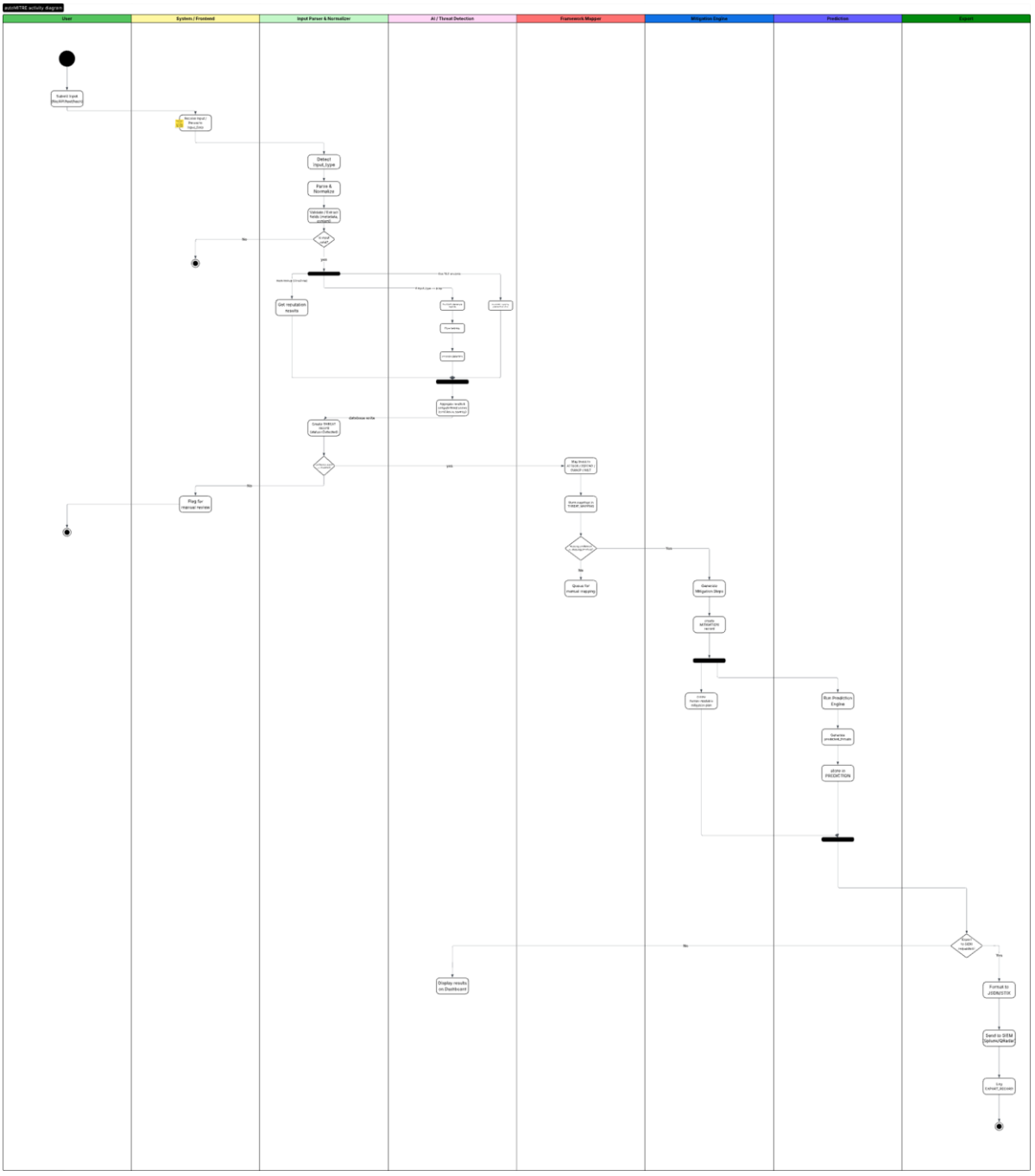
5.2.2 Use Case Diagram:

Figure 2: Use Case Diagram



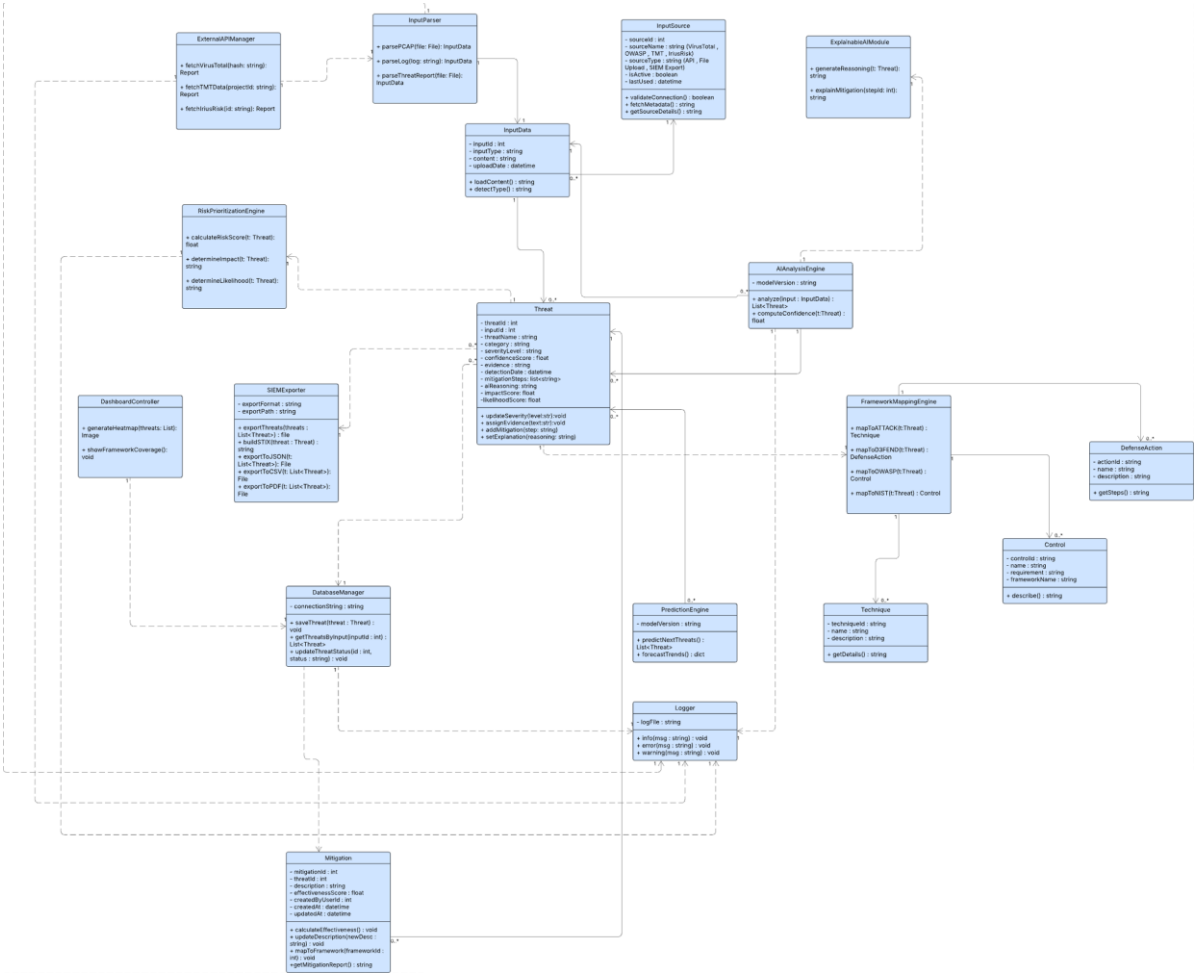
5.2.3 Activity Diagram:

Figure 1: Activity Diagram



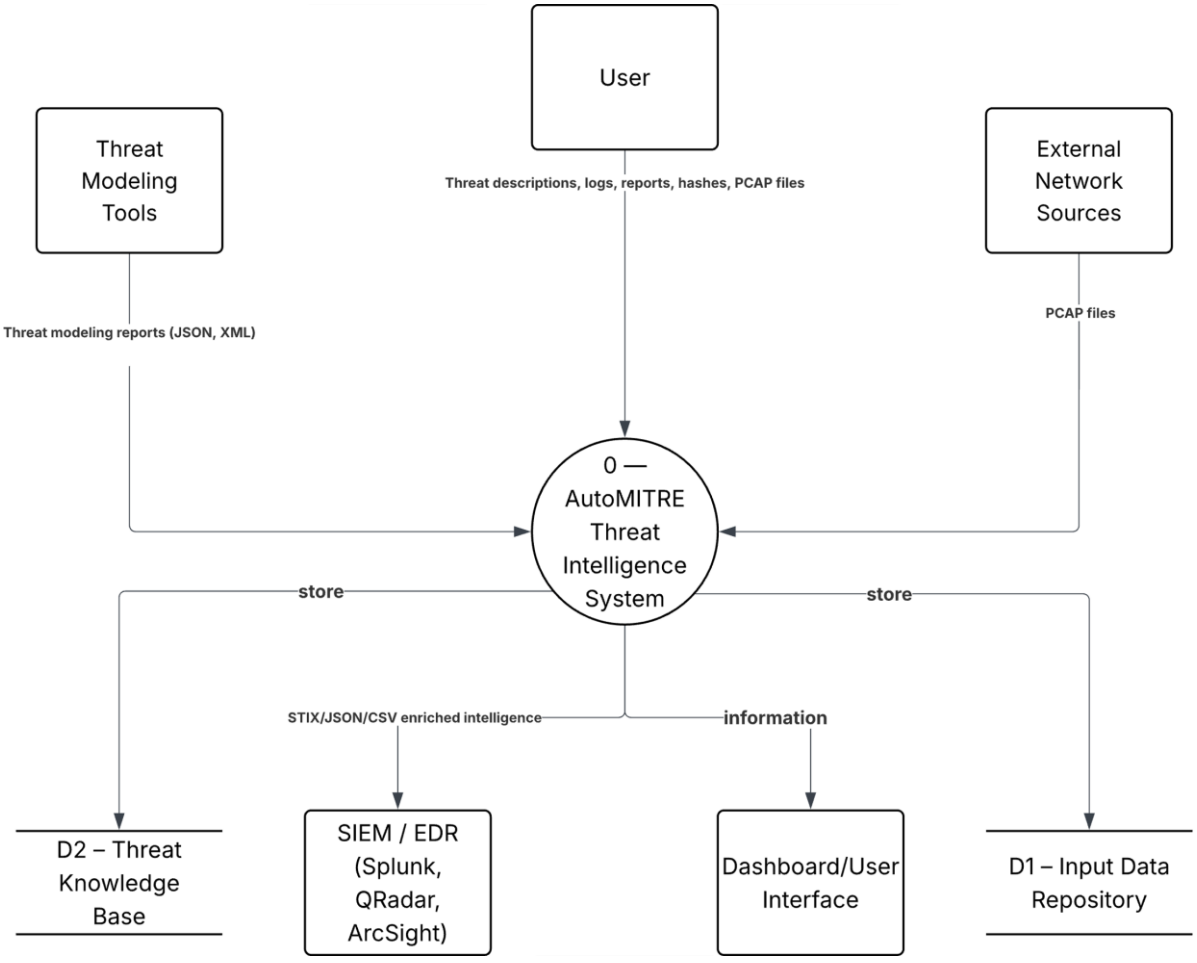
5.2.4 Class Diagram:

Figure 2: Class diagram



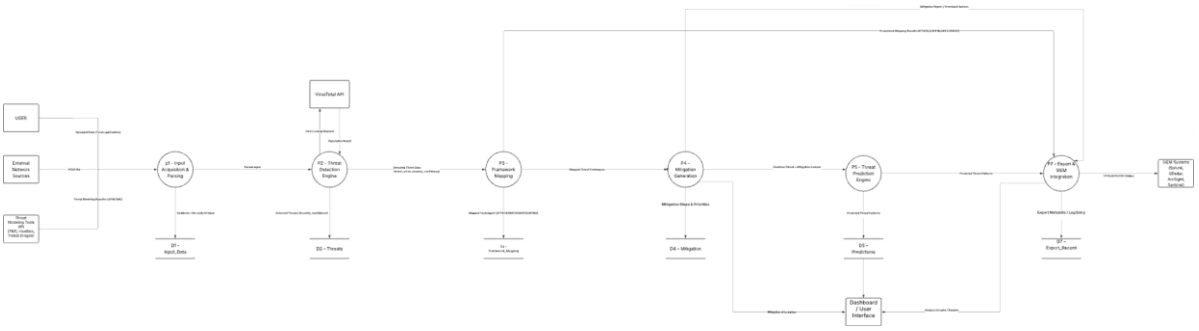
5.2.5 Data Flow Diagram level 0 (DFD 0)

Figure 3: Data Flow Diagram Level 0



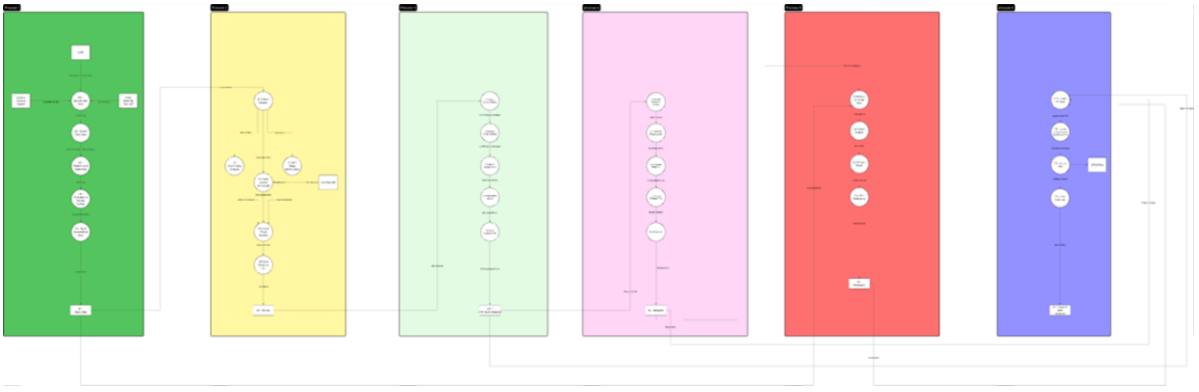
5.2.6 Data Flow Diagram level 1 (DFD 1):

Figure 4: Data Flow Diagram Level 1



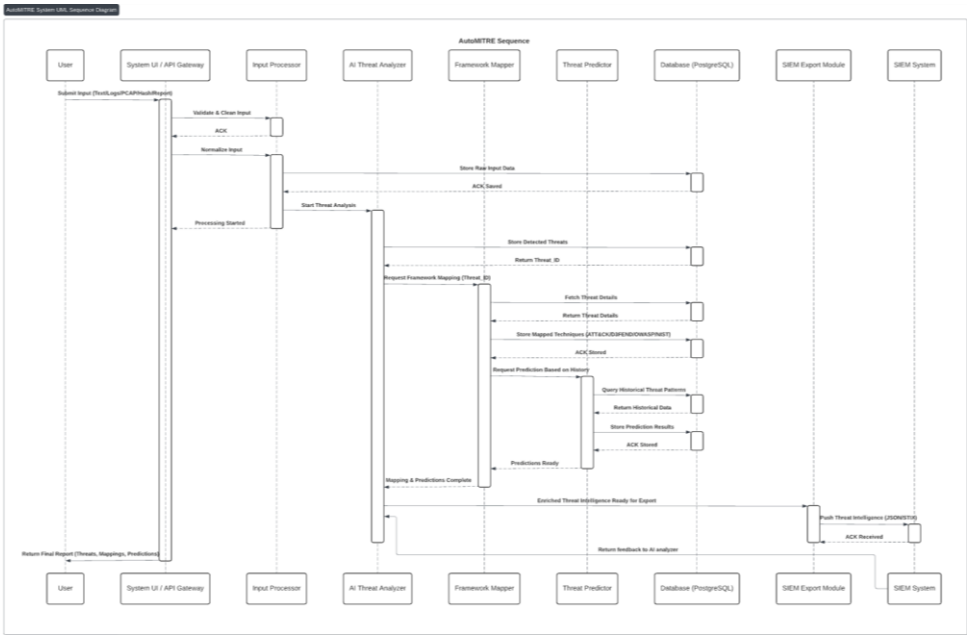
5.2.7 Data Flow Diagram level 2 (DFD 2):

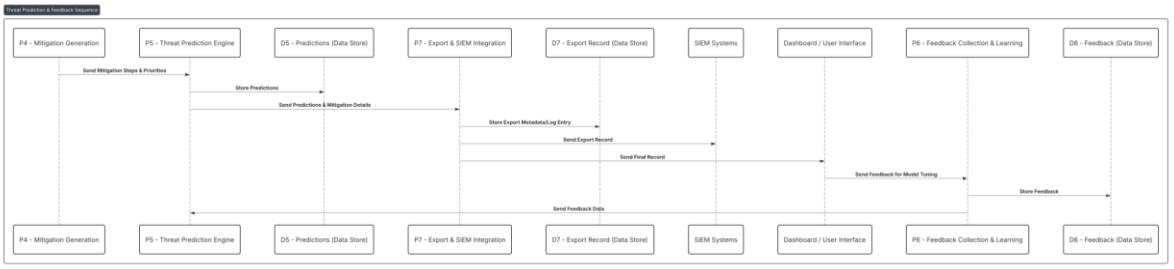
Figure 5: Data Flow Diagram Level 2

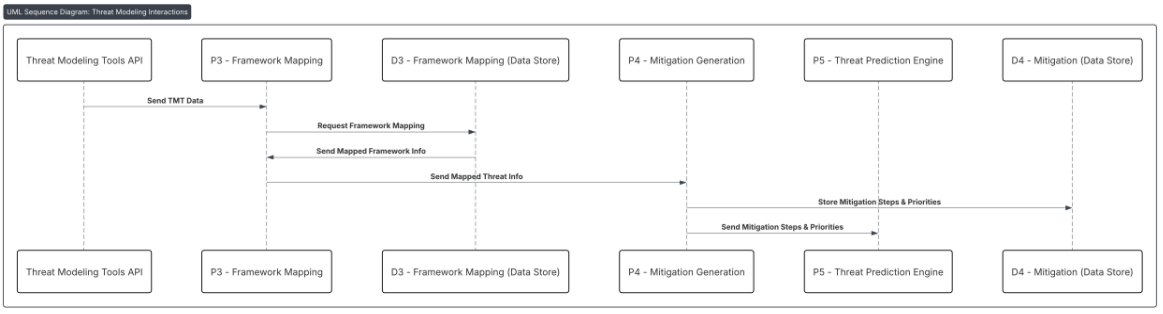


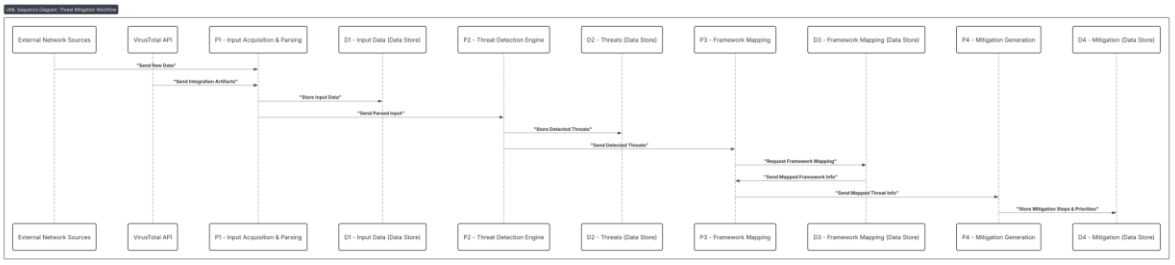
5.2.8 Sequence Diagram:

Figure 6: Sequence Diagram



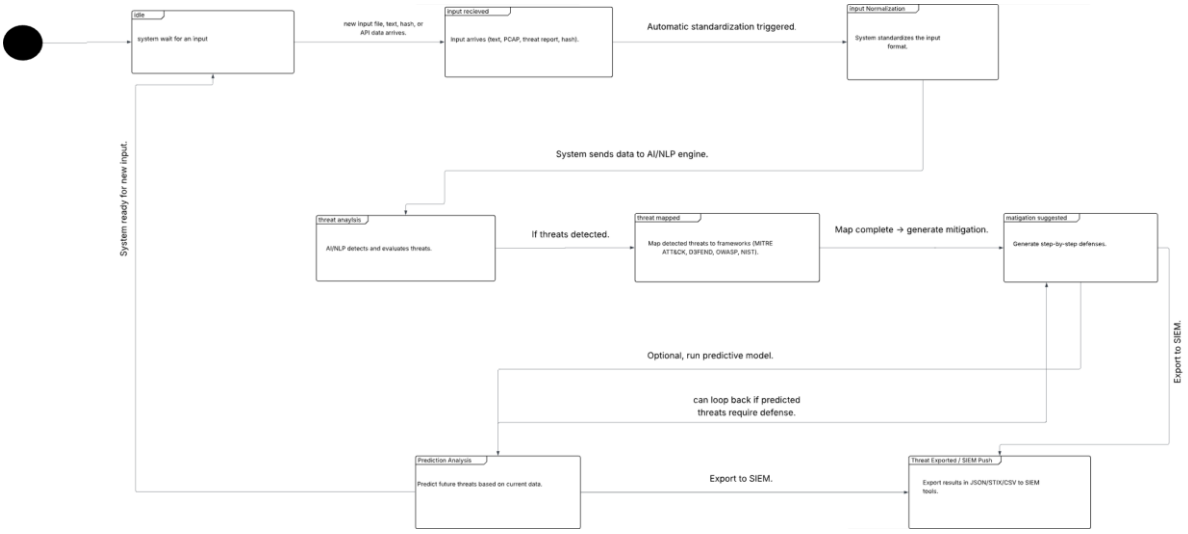






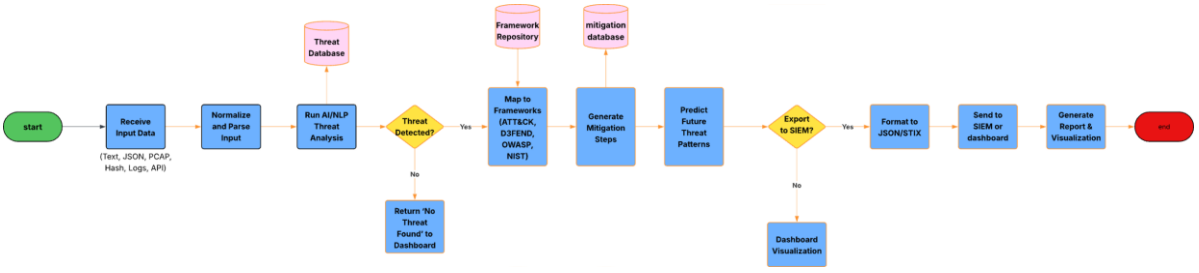
5.2.9 State Machine Diagram:

Figure 7: State Machine Diagram



5.2.10 Flowchart Diagram :

Figure 8: Flowchart Diagram



5.3 Technologies and Tools Used

This section describes the technologies, tools, and frameworks used in the design and implementation of the **AutoMITRE** system.

5.3.1 Programming Languages

- **Python**
  Used as the primary backend language for threat analysis, AI processing, framework mapping, and system logic due to its rich ecosystem of cybersecurity and machine learning libraries.
- **JavaScript**
  Used for frontend development to build interactive user interfaces and dashboards.

5.3.2 Backend Frameworks

- **FastAPI / Flask**
  Used to develop RESTful APIs that handle user requests, file uploads (PCAP, logs, reports), integration with external tools, and communication between system modules.

5.3.3 Artificial Intelligence & Machine Learning

- **Scikit-learn**
  Used for implementing machine learning models for threat classification and pattern detection.
- **Natural Language Processing (NLP) Libraries**
  - spaCy

    - NLTK
      Used to analyze text-based threat descriptions and threat modeling reports.
- **Rule-Based + ML Hybrid Approach**
  Combines predefined security rules with machine learning models to improve accuracy and explainability.

5.3.4 Cybersecurity Frameworks

- **MITRE ATT&CK**
  Used for mapping detected threats to known attacker techniques.
- **MITRE D3FEND**
  Used to recommend defensive techniques corresponding to detected threats.
- **OWASP (ASVS / Top 10)**
  Used for application-level security mapping.
- **NIST SP 800-53**
  Used for mapping threats to security controls and compliance requirements.

5.3.5 External APIs & Integrations

- **VirusTotal API**
  Used to analyze malware hashes and retrieve threat intelligence.
- **Threat Modeling Tools APIs**
    - Microsoft Threat Modeling Tool (TMT)
    - OWASP Threat Dragon
    - IriusRisk
      Used to retrieve generated threat modeling reports directly through API integration.
- **SIEM Integration**
  Supports exporting analysis results in **JSON, STIX, and CSV** formats to be consumed by SIEM platforms such as Splunk, QRadar, and ArcSight.

5.3.6 Data Processing & File Handling

- **PCAP Analysis Tools**
  Used for parsing and analyzing network traffic data.

- **Log Processing Modules**
  Used to analyze system, network, and security logs from SIEM or EDR sources.

5.3.7 Frontend & Visualization

- **React.js**
  Used to build the user interface and dashboards.
- **Data Visualization Libraries**
  Used to generate risk heatmaps, charts, and framework coverage visualizations.

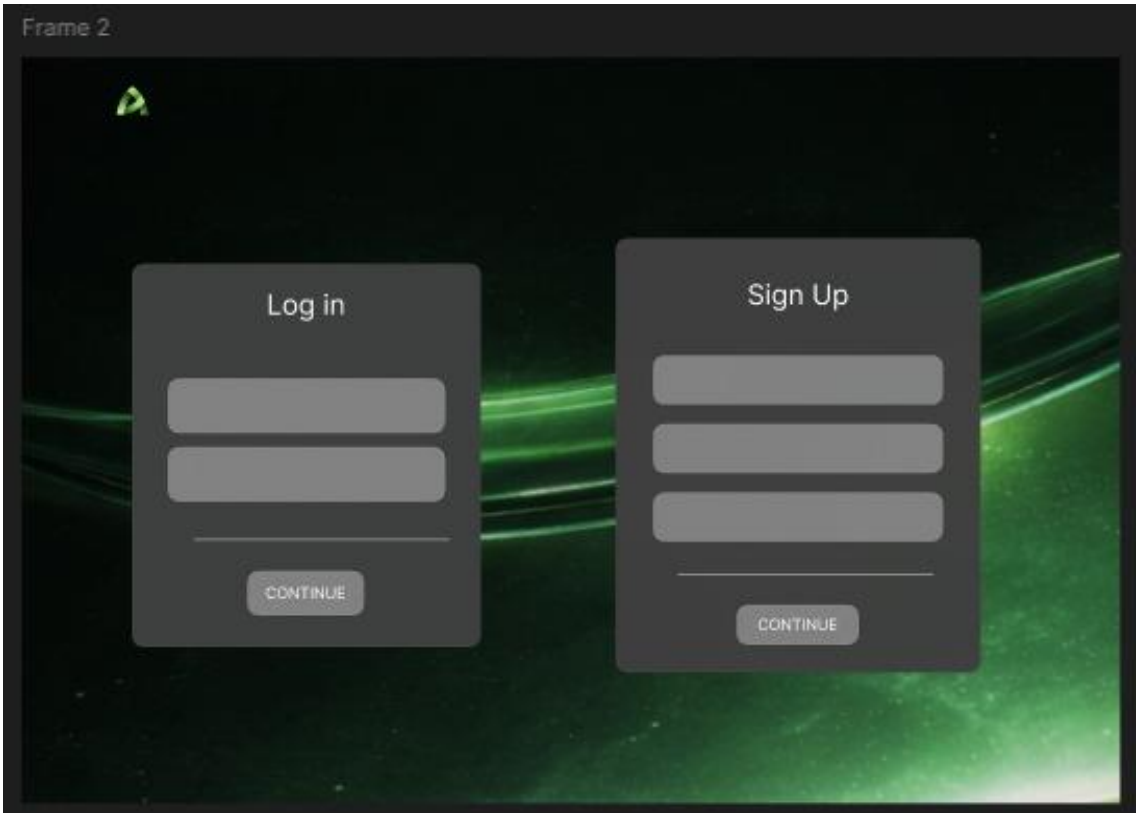
5.3.8 Design & Prototyping Tools

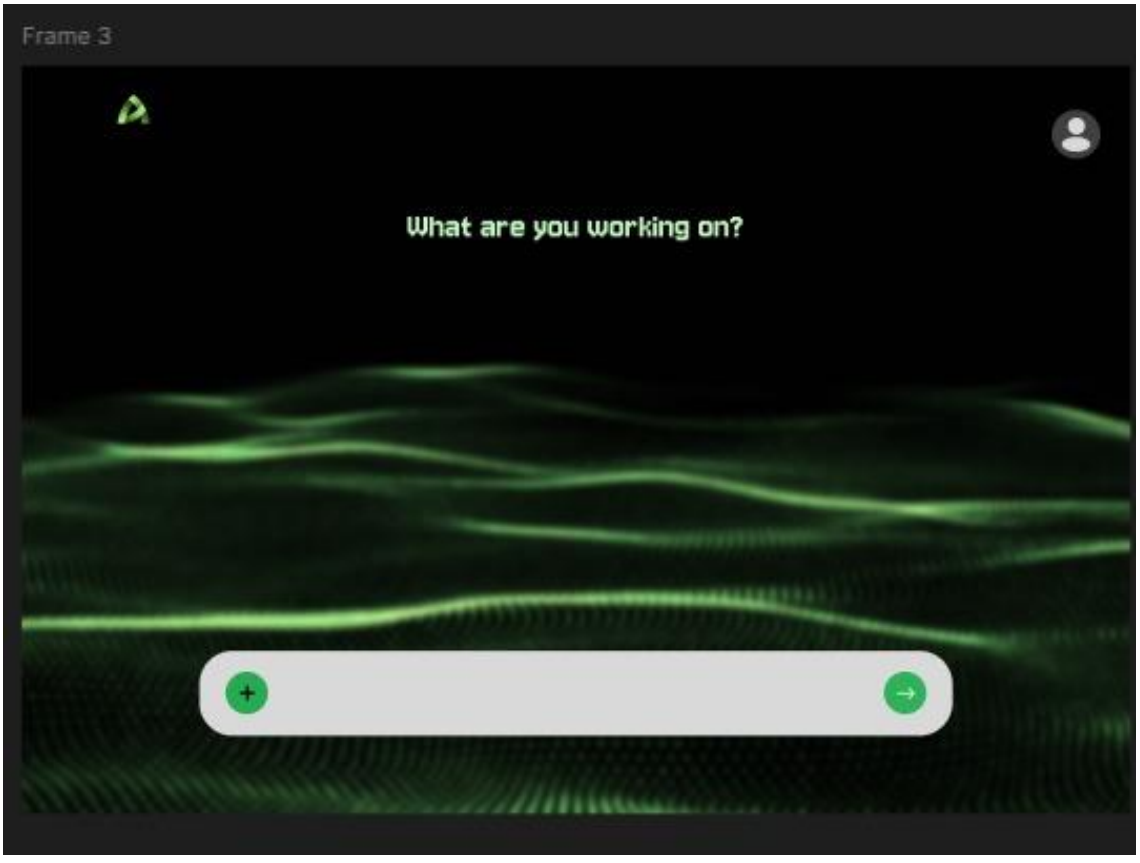
- **Figma**
  Used for UI/UX prototyping and interface design.
- **Draw.io / Lucidchart**
  Used for designing system architecture diagrams and UML diagrams.

5.3.9 Development & Deployment Environment

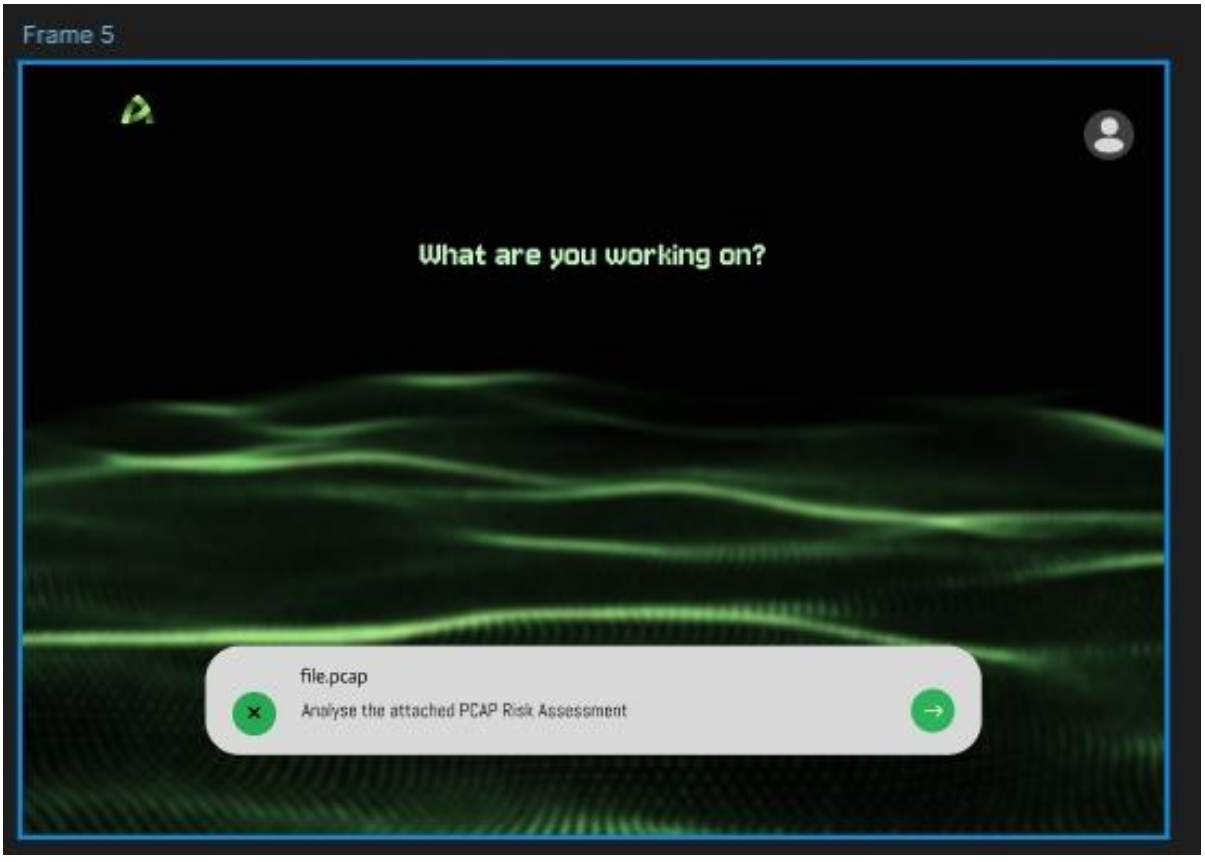
- **Git & GitHub**
  Used for version control and team collaboration.
- **Localhost / Cloud Free Tier**

5.4 Prototype


Frame 2
Log in
CONTINUE
Sign Up
CONTINUE


Frame 3
What are you working on?

Frame 4
What are you working on?
Threat Model
PCAP
Malware Hash
Logs


Frame 5
What are you working on?
file.pcap
Analyse the attached PCAP Risk Assessment

Frame 6

file.pcap

Analyse the attached PCAP Risk Assessment

Risk Assessment

- Severity: HIGH
- Likely scenario: Infected internal endpoint communicating with external C2 infrastructure
- Assets at risk: Internal network, credentials, sensitive data
- Business impact: Data exposure, lateral compromise, operational disruption

Frame 7

User Name: Mira Amin

What are you working on?

Chapter 6 - Implementation

Add introductory paragraph.

1. Implementation Environment

6 Add sections as required.

7 Results and Discussion

Chapter 7 - Testing

Software Test Plan IEEE 829

Chapter 8 - Conclusion and Future Work

Add conclusion and future work, they can be separated in two sections or paragraphs.

References

Appendix A - Senior Project Summary Report

Project Title		
Supervisor(s)		
Team members:	**Names** …… …… …… …… ……	**Registration Numbers** …… …… …… …… ……
Project Deliverables	*Indicate the deliverables of the project.*	
Local Impact	*Briefly state the local impact of the project on the following aspects ; social, economical, environmental, any other relevant areas)*	
Global Impact	*Briefly state the global impact of the project on the following aspects ; social, economical, environmental, any other relevant areas)*	
Team Organization	*Describe the interdisciplinary/multidisciplinary team organization; list each member role and contributions. How each member evaluates a problem and performs duties within a team environment.*	
Ethical Considerations	*Describe any ethical and/or safety issues related to the project (if exits)*	
Social Impact	*Describe the social impact of the project.*	
Professional Responsibility	*Illustrate how professional responsibility was taken into consideration while working on your project.*	
Standards	*List the standards incorporated in your project*	
Realistic Constraints	*List the realistic constraints you considered in your project*	

Supervisor Name	Signature